Exercise 1 — Rank, systems, and Rouché–Capelli

Consider the system in x, y, z, with parameters $a, b \in \mathbb{R}$:

$$\begin{aligned} x + 2y + z &= 1 \\ 2x + 4y + az &= 2 \\ x + ay + z &= b \end{aligned}$$

(a) Compute $\det(A)$ for the coefficient matrix and determine for which a the matrix is invertible.
(b) Using Gauss–Jordan on the augmented matrix, classify the solution set for every pair (a, b).
(c) In the case with infinitely many solutions, give the dimension of the solution set and an explicit parametrization.

Exercise 1 — Solution

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Exercise 2 — Subspaces, span, and basis

Work in $V = M_{2\times 2}(\mathbb{R})$ with the usual matrix addition and scalar multiplication (you may assume this is a vector space).

(a) Let $W = \{A \in V : \mathrm{tr}(A) = 0\}$. Verify the three subspace conditions and produce a basis. What is $\dim W$?

(b) Let $S = \{A \in V : A = A^\top\}$. Show S is a subspace and find $\dim S$.

(c) Find a basis for $W \cap S$. (Any intersection of vector subspaces is a vector subspace.)

(d) Let $U = \{A \in V : A \text{ is invertible}\}$. Show U is *not* a subspace in two different ways.

Exercise 2 — Solution

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Exercise 3 — Diagonalization and a linear dynamic system

Let

$$A = \begin{pmatrix} 0.5 & 0.3 \\ 0.2 & 0.6 \end{pmatrix}, \qquad x_t = Ax_{t-1}, \qquad x_0 = \begin{pmatrix} 5 \\ 0 \end{pmatrix}.$$

(a) Find the characteristic polynomial and the spectrum. Verify Proposition 12: $\det A = \prod_k \lambda_k^{a_k}$ and $\operatorname{tr} A = \sum_k a_k \lambda_k$.

(b) Find eigenvectors, construct P and Λ with $A = P\Lambda P^{-1}$, and derive a closed form for x_t. What is $\lim_{t\to\infty} x_t$, and which eigenvalue governs the rate of convergence?

(c) Now let $B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Show B is not diagonalizable by comparing algebraic and geometric multiplicity. Does $B^t x_0 \to 0$? What does this say about the converse of the stability argument in **(b)**?

Exercise 3 — Solution

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Exercise 4 — Definiteness, Sylvester, and a caveat

Let

$$A(a) = \begin{pmatrix} 2 & 1 & 0 \\ 1 & a & 1 \\ 0 & 1 & 2 \end{pmatrix}, \qquad a \in \mathbb{R}.$$

(a) Use Sylvester's criterion to find all a for which $A(a)$ is positive definite.

(b) At the critical value of a, show A is positive semi-definite and find a nonzero x with $x^\top Ax = 0$. Relate this vector to the spectrum of A.

(c) Suppose $A(a)$ is the Hessian of $f : \mathbb{R}^3 \to \mathbb{R}$ at a critical point. For which a is that point a strict local minimum? A strict local maximum?

(d) Consider $M = \left(\begin{smallmatrix} 0 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Compute its *leading* principal minors and check them against clause 3 of Theorem 9 on slide 50. Then compute $x^\top Mx$ directly. What is the correct semi-definiteness criterion?

Exercise 4 — Solution

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